Box 15.1: Repo Rate and Bank Rate

Most of the time we are confused between various interest rates such as bank rate or discount rate, repo rate and neighbouring rate in the banking system.

Repo Rate: The banks approach the central bank for financial assistance if they face crisis. The interest rate charged on banks by RBI to provide such financial assistance against of certain securities or bonds, is the repo rate. In other words, "commercial banks borrow money from RBI by selling securities or bonds with an agreement to repurchase the securities on a certain date at a predetermined price. The rate of interest charged by the central bank on the cash borrowed by commercial banks is called the Repo Rate". For example, if a commercial bank borrows or takes a loan of Rs. 100,000 at the predetermined repo rate of 5%, the interest to be paid to RBI at the maturity is Rs.5000.

In other scenario, if a commercial bank has excess funds, the bank deposits it in RBI and earns interest at a rate known as 'reverse repo rate'. For example, if a commercial bank deposits Rs. 100,000 at the reverse repo rate of 4%, the interest to be paid by RBI to the bank at the maturity is Rs.4000.

RBI controls the liquidity in the market through repo rate. With an objective of increasing liquidity, RBI reduces repo rate, which encourages the commercial banks to sell their securities and borrow from the RBI. On the other hand, RBI increases the repo rate, which discourages the banks from selling their securities, which decreases liquidity or the supply of money.

Bank Rate: The rate of interest charged by the central bank on the loans it has extended to commercial banks and other financial institutions is called "Bank Rate". In this case, there is no repurchasing agreement signed, no securities sold, and no collateral involved.

Banks also borrow funds from RBI and extend loan to their customers, which is an important source of their profit. Hence, they borrow funds from RBI without involvement of the repurchasing agreement, securities or bonds. The interest charged on such borrowing of commercial bank is the bank rate, which is usually higher than Repo Rate. This is an important weapon of RBI to control liquidity in the market. The increase in bank rate by RBI increases the borrowing costs of the banks which, in turn, reduce the supply of money. The bank rate is also called the discount rate.

There is another concept in the banking system that is the overnight rate. The commercial banks also borrow funds among themselves. The rate charged on such borrowing of funds is known as the overnight rate.

Sometimes the OMO is used to neutralise the impact of other factors on money supply. For instance, change in the exchange rate affects money supply. The central bank intervenes in the foreign exchange market by buying or selling foreign exchange to regulate the volatility of the domestic currency. Suppose, for example, the objective of the RBI is to check the appreciation of the Rupee. In order to realise this objective, the RBI would buy foreign currency (say US Dollar). With supply remaining the same, if the demand for US Dollar increases, the value of Indian Rupee will decrease. In the process, there would be some increase in money supply. In order to neutralise the increase in money supply, the RBI may choose the option of open market operations, and sell government securities in the market. It means that a central bank sells government securities to absorb the extra money supply, which came up due to its exchange rate policies. Neutralising such impact of change in foreign exchange reserves on the money supply through the OMO, or the selling of government securities, is known as the process of 'sterilization'.

Specifically, if the goal is to control inflation, the central bank sets a target of a higher interest rate by selling securities in the open market. A higher interest rate indicates less liquidity in the banking system, which leads to an increase in the interest rates charged by banks to their customers. Along with the decline in consumer expenditure, it also makes certain investment projects costly. As a consequence of higher interest rate, investment spending in the economy will decline. As a result of higher interest rate, consumption expenditure may decline, particularly those segment which is funded by borrowings from commercial banks. Further, households may allocate more to saving and less to consumption because of the increase in interest rate. As these are the components of aggregate demand, aggregate demand would decline because of the rise in interest rate. If aggregate demand falls, the price level would be negatively affected and the intended goal of controlling inflation is realised.

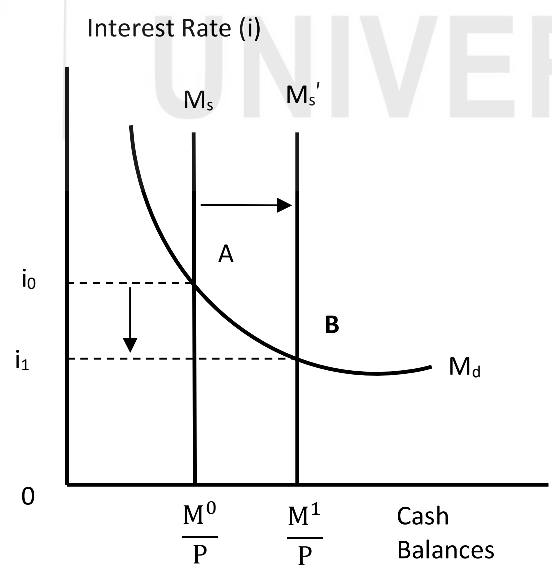

Fig.15.1: Effect of Increase in Money Supply

Let us explain the effect of an increase in interest rate on money supply through diagram. In Fig. 15.1 we indicate cash balances, money supply and money demand on the x-axis. On the y-axis we take interest rate (*i*). We know that demand for money depends upon income and interest rate such that, Md = f[Y, i]. Here demand for money is an increasing function in income (Y) and decreasing function in interest rate. Money supply is given by a vertical straight line at the level Ms = M/P, where M/P indicates the *purchasing power* of available money balances. In equilibrium,

$$M_d = M_s \text{ or } M_d = M/P \qquad \ldots(15.8)$$

If the price level remains unchanged, the real cash balances can be changed only through the changes in money supply. An increase in money supply or real money balances through OMO tools of the central bank shifts the M_S function to the right (from M_s to M_s' in Fig.15.1). The main goal of such a policy is to stimulate aggregate demand. The implementation of this policy results in excess supply of money mainly in the form of excess reserves and thus loanable funds/ reserves. Consequently, it results in a fall in the interest rates charged by the commercial banks to customers on credit.

Check Your Progress 3

1) What is meant by open market operations? Explain how it affects money supply and interest rate in the economy.

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2) Explain the differences between bank rate, discount rate, repo rate, reverse repo rate, and overnight rate.

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15.4 LET US SUM UP

There is no unique definition of money supply. Central banks measure money supply through a series of monetary measures. The currency notes and coins, and the deposits of banks with the central bank form the monetary base of an economy. Monetary aggregates such as M0 and M1 are known as 'narrow money' whereas M3 is said to be 'broad money'.

The monetary base of an economy increases over time, which increases the money stock in the economy in a multiplicative manner. The multiple by which money stock increases against the monetary base is known as money multiplier. Money supply can be controlled through various tools of the central bank such as open market operations, changes in the rate of interests and exchange rates. If the objective of the central bank is to check inflation, the central bank would regulate money supply by buying government securities from the commercial banks and other financial institutions. The RBI can also increase the bank rate and repo rate, which will increase the borrowing cost of commercial banks, which in turn would decrease money supply. Higher interest rate affects investments and household consumptions negatively. It leads to decline in the aggregate demand in the economy which affects the price level. That means there is a strong linkage between the monetary policy of an economy and its real sector.

15.5 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Refer to Section 15.2 and Table 15.1. Your answer should include the monetary aggregates NM0 to NM3, and L1 to L3.
2) Refer to Section 15.2, particularly Table 15.1, and answer.
3) Refer to Section 15.2, particularly Table 15.1, and answer.

Check Your Progress 2

1) It is the ratio of the money stock measured as M1 to the monetary base. Derive the equations on the basis of Section 15.3.
2) It denotes the ratio of currency with the non-bank public to the money stock measured by M1. Derive equation (15.6) and interpret it.

Check Your Progress 3

1) The process of changing the monetary base through buying or selling of the government securities is known as the Open Market Operations. Use Fig.15.1 to explain the effect of interest rate on money supply and interest rate.
2) Refer to Box 15.1 in Section 15.3 and answer.

UNIT 16 CONDUCT OF MONETARY POLICY*

Structure

16.0 OBJECTIVES

After going through the unit you will be in a position to:

- distinguish between goals and targets of monetary policy;
- explain various credit control measures under monetary policy;
- identify the instruments of monetary policy and their role; and
- identify and explain the monetary transmission channels.

16.1 INTRODUCTION

In Unit 5 of this course we learnt that in an open economy with flexible exchange rate, monetary policy is effective in bringing about economic growth. You can appreciate the importance of monetary policy in view of the fact that most economies these days follow a flexible or floating exchange rate.

You may recall that the Classical Economists assumed money to be neutral in the sense that it does not have any effect on prices and economic growth. They believed that there is duality in the economy such that monetary and real variables do not influence one another. It was Keynes who suggested that money supply affects real variables. According to him increase in money supply leads to a decline in rate of interest, which in turn leads to greater borrowings by firms thereby leading higher investment. Higher investment leads to higher output and income of the country through a multiplier effect. Thus he visualized a greater role for the government. According to him economic policy should be countercyclical – it should neutralize the effects of business cycles. The New Classical economists attempted to justify the Classical views on the neutrality of

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money and brought forth the 'policy ineffectiveness proposition'. According to the New Classical economists, it is best for the government to stick to certain 'policy rules'. Governments in all the countries, however, see an important role for monetary policy in the economy and actively pursue a monetary policy.

In this Unit we will discuss the types of monetary policy that a government pursues, the objectives of monetary policy, and the policy instruments available to policy makers. In addition, we will discuss how monetary policy works. We will bring out certain important channels; the effect of an increase in money supply on other economic variables.

16.2 MONETARY POLICY TYPES

The macroeconomic policy set by the central bank or the monetary authority of a country is known as monetary policy. It is concerned with the management of money supply in an economy. The change in money supply usually results in a change in rate of interest. The objective of monetary policy is to fulfill the objectives of inflation, growth, consumption and liquidity in the economy. Monetary policy also aims at maintaining exchange rates with other currencies in order to benefit trade, international confidence and policy stability in the country. There are certain instruments of monetary policy as we will learn later in this Unit.

Monetary policy could be of two types: expansionary and contractionary. With an expansionary monetary policy, the monetary authority tries to stimulate the economy using the policy instruments. In contrast, the monetary authority attempts to restrict money supply and control inflation with a contractionary monetary policy. In an expansionary monetary policy, either a lower than usual rate of interest is maintained or money supply is abnormally increased. This is done with the hope of providing cheaper credit to the firms so that investment in the economy increases. Increase in investment is expected to further increase employment and output. Expansionary monetary policy, however, often leads to deterioration of the exchange rate of the country with other currencies. If increases in money supply lead to higher inflation in the economy, depreciation of the domestic currency in the foreign exchange market may take place.

Under contractionary monetary policy, the instruments of monetary policy aim at a different set of results. If the economy is passing through a very high inflationary condition (if the economy is over-heated), the monetary authority usually resorts to a higher than usual rate of interest or abnormally low money supply. If contractionary monetary policy is implemented too vigorously, there may be decrease in production, increase in unemployment, and fall in aggregate demand leading to recessionary stage of the business cycle.

16.3 GOALS AND TARGETS OF MONETARY POLICY

You should note that there is a difference between the *goals* and the *targets* of monetary policy. Goals of monetary policy are the objectives it aims to achieve, that is, reducing inflation, achieving full employment or maximising economic growth. On the other hand, targets of monetary policy are the variables targeted such as money supply, bank credit, and short term interest rates through the instruments of monetary policy.

Monetary policy is one of the instruments of the overall economic policy of a country; hence its objectives are no different than the overall economic policy objectives of the economy. As per the prevailing economic, political, historical and international conditions, there are changes in the priorities given to the objectives of the monetary policy. Hence, they differ amongst developed and developing countries. The significant objectives of monetary policy are

- to ensure price stability, that is, containing inflation,
- to promote economic growth,
- to ensure stability of the exchange rate of the currency,
- to achieve full employment,
- to reach equilibrium in the balance of payments account, and
- to reduce economic inequality.

Let us explain below these objectives in some detail.

a) Price Stability

Controlling inflation is generally considered to be the foremost objective amongst developing economies as high inflation has an otherwise negative impact on growth and distribution. Price stability means relative price stability, depending on the prevailing conditions in the economy. Hence, by maintaining relative price stability there is a considerable check on the fluctuations in the business cycle leading to stability in the economic growth process as a whole.

Price stability does not mean that prices should not increase at all. There should be a reasonable rate of inflation in an economy so that there is incentive for production and economic growth. There is no consensus among economists, however, on an ideal rate of inflation; it varies across countries and over period of time. In countries like the US, an inflation rate of around 2 per cent is targeted. In India, an inflation rate of 4 per cent with a tolerance band of 2 per cent is considered as ideal (it implies that inflation should not be outside the range of 2 per cent to 6 per cent).

Many countries have faced episodes of abnormally high inflation. In 2019, for example, some countries are experiencing exceptionally high inflation – Venezuela has an inflation rate of about 10 million per cent per year! Argentina, Iran and Sudan are facing inflation rate of above 25 per cent per year. Abnormally high inflation leads to rise in the cost of living of the poor, makes exports more expensive, lowers the rate of savings which further hampers economic growth and encourages unproductive investment such as jewelry, real estate, etc. During abnormally high inflationary scenario, contractionary monetary policy is used. In period of relatively stable inflation, the monetary authority pursues a monetary policy such that price stability with economic growth and employment generation are maintained.

b) Economic Growth

Promoting economic growth is another important objective of both developed and developing countries. Economic growth implies growth in the productive capacity of the economy so that real national income increases. To achieve economic growth, the rate of savings and investments need to increase. Monetary policy can help produce favorable effects to the requirements of economic growth by directing the banking sector towards priority sector lending or making appropriate changes in interest rate to incentivize people to save.

With the openness of the economy and floating exchange rate, the goal to achieve increased economic growth through instruments of monetary policy may contradict with the goal of exchange rate stability. In order to prevent the depreciation of the currency, a tight monetary policy need to be maintained but the higher rate of interests will reduce liquidity in the banking system. This reduction in liquidity results in increased cost of lending and lesser production and increased unemployment in the economy. These contradictory objectives result in a trade off in the policy circles and considering the other political, social and economic objectives, a final decision is reached.

c) Exchange Rate Stability

Almost all major economies of the world have started following the flexible exchange rate system under which the exchange rate of the currency is determined by the demand for and supply of foreign exchange. Whenever there is disequilibrium in the demand and supply of foreign exchange, the external value of the currency changes. The domestic investors demand foreign currency when there is profitable investment possible abroad whereas they demand home currency more when there is profitable investment at home.

Under floating exchange rate the central bank or the monetary authority intervenes to avoid sharp fluctuations in the exchange rate. In order to prevent depreciation of rupee, the central bank can release more foreign currency from its foreign exchange reserves. Also the central bank can increase the short term lending rates and increase the required cash reserve ratio (CRR) in order to reduce the demand for the foreign currency.

d) Full Employment

In order to achieve maximum social welfare it is imperative that the economy attains full employment. Any country will prosper when all its people get the job that they desire to do at the prevailing wage rate. With a tight monetary policy, the objective of full employment and growth contradicts. Hence, in order to increase the level of employment the central bank or the monetary authority will have to adopt expansionary monetary policy.

e) Balance of Payments Equilibrium

Abnormally high current account deficit (CAD) is financed through increase in nominal capital inflows which is achieved through the depreciation of the home currency. Thus, imports become very expensive and there is increase in inflation in the economy. A tighter monetary policy helps reduce these effects through high cost of credit and lower availability of money with the importers, thus curbing imports.

Hence, monetary policy plays a significant role in maintaining the current account deficit at the reasonable level. Balance of Payments equilibrium implies exchange rate stability.

f) Economic Inequality

There is increased income inequality amongst the developed and developing countries. This increased inequality leads to reduction in social welfare and increased frustration amongst the masses. Monetary policy as an economic instrument aims to achieve equitable distribution of income and wealth through priority sector availability of credit to the weaker sections of the society at very low rate of interests.

The targets of monetary policy are generally the following three parameters:

- to regulate the supply of money/ credit
- to maintain a reasonable rate of interest or cost of money/credit
- to conveniently avail money/ credit

a) Supply of Money

The currency issued by the monetary authority and the demand deposits with the commercial bank comprise the supply of money. The issue of currency by the central bank needs to be flexible enough depending on the inflationary or recessionary stages prevailing in the economy. The banking system also needs to play a progressive role in facilitating demand deposits with the banks.

b) Rate of interest or Cost of Money

Underdeveloped economies generally adopt an expansionary monetary policy in order to promote sectors like agriculture and industry. Policy of discriminatory rate of interests is often followed in order to promote priority sector lending. For example, lower rate of interest is charged from the priority sectors such as micro, small and medium enterprises (MSME). Simultaneously, higher rates of interest may be charged from unproductive and luxurious spending so that higher credit is made available to the priority sector.

During inflationary stage of the business cycle, the monetary authority keeps the rate of interest at a higher level to curb money supply in the economy. In contrast, during the recession phase, the monetary authority may decide to keep the interest rate at a lower rate. It is expected that lower interest rate will provide incentives for higher production, employment and growth.